Fisher Markets

Convex Programming and Duality

Zhiyun Guo

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1 Preliminaries

1.1 Notation

- M: a set of m divisible goods
- N: a set of n buyers
- B_i: Each buyer i's budget (fixed)
- $p_j \in \mathbb{R}_+$: price of good j
- $x_{ij} \in \mathbb{R}_+$: amount of good j allocated to buyer i

Standing assumptions. Budgets satisfy $B_i > 0$. Goods are divisible and have unit supply after rescaling positive supplies. For the linear-market programs below, each buyer values some good, and each good is valued by some buyer. Nonnegativity is allowed for v_{ij}; zero-valued pairs are excluded when taking logarithms or ratios.

1.2 Utility

1. *Linear*: there are parameters $v_{ij} \in \mathbb{R}_+$ specifying the value derived by buyer i from one unit of good j which guarantees **Additivity**. i.e., $u_i(x) = \sum_{j \in M} v_{ij} x_{ij}$.
2. *Quasi-linear*: agents have utility for the money spent as well, i.e., $u_i(x) = \sum_{j \in M} (v_{ij} - p_j) x_{ij}$.
3. *Leontief*: given coefficients $a_{ij} \geq 0$, with at least one positive coefficient per buyer, $u_i(x) = \min_{j : a_{ij} > 0} x_{ij}/a_{ij}$. Goods with $a_{ij} = 0$ are not required inputs.
4. *Constant elasticity of substitution (CES)*: for example, when $0 < \rho < 1$,

$$u_i(x_i) = \left(\sum_j \alpha_{ij} x_{ij}^{\rho} \right)^{1/\rho},$$

with $\alpha_{ij} \geq 0$ and some positive coefficient. Other parameter regimes, including complementary goods and the Cobb–Douglas limit, require the corresponding domain/limit conventions.

2 Market equilibrium

An *equilibrium* is formed if the following conditions hold.

1. The allocation of each buyer i **maximizes his utility**, subject to the budget constraint $\sum_j p_j x_{ij} \le B_i$
2. Each good j that has a price $p_j > 0$ is **allocated fully**, i.e., $\sum_i x_{ij} = 1$. A good is allowed to have $p_j = 0$ as long as $\sum_i x_{ij} \le 1$.

2.1 Generalization of Fisher's model

2.1.1 Spending-Restricted (SR) model

- Each seller j has an upper bound on the amount of money he wants to earn: c_j
- Once he earns c_j, selling the least amount of his good, he wants to take back the unsold portion of his good, i.e., the amount of money spent on the good of seller j is restricted by c_j
- In equilibrium, the second equilibrium condition above is modified to $\forall j \in M, \sum_i x_{ij} \le 1, \sum_i p_j x_{ij} \le c_j$, and either

$$\sum_i x_{ij} = 1 \text{ or } \sum_i p_j x_{ij} = c_j, \text{ or } p_j = 0.$$

2.1.2 Utility-Restricted (UR) model

- Each buyer i has an upper bound on the utility they want to derive in the market: d_i
- Once buyer i derives utility d_i, spending the least amount of money at prices p, she wants to keep the left-over money. i.e., the utility of buyer i is restricted by d_i
- The first equilibrium condition is modified to $\forall i \in N,\ u_i(x) \le d_i$, and $\sum_j p_j x_{ij} \le B_i$, and either

$$x \text{ minimizes } \sum_j p_j x_{ij} \ s.t.\ u_i(x) = d_i, \text{ or maximizes } u_i(x) \ s.t. \sum_j p_j x_{ij} \le B_i.$$

3 Convex programming duality

3.1 Fenchel Conjugate

Definition 3.1 *Fenchel conjugate*

For $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$, its conjugate is

$$f^*(\mu) = \sup_{x \in \operatorname{dom} f} \{\mu^\top x - f(x)\}.$$

The conjugate may be extended-valued. It is convex as a supremum of affine functions; strict convexity and differentiability do not follow in general.

Useful rules and assumptions.

- If f is proper, closed and convex, $f^{**} = f$ (Fenchel–Moreau).
- If $f(x) = \sum_i f_i(x_i)$ on a product domain, $f^*(\mu) = \sum_i f_i^*(\mu_i)$.

- If $g = cf$ with $c > 0$, then $g^*(\mu) = cf^*(\mu/c)$.
- If $g(x) = f(cx)$ with scalar $c \neq 0$, then $g^*(\mu) = f^*(\mu/c)$.
- If $g(x) = f(x + a)$, then $g^*(\mu) = f^*(\mu) - \mu^\top a$.
- For proper closed convex f, Fenchel equality is equivalent to a subgradient pair:

$$f(x) + f^*(\mu) = \mu^\top x \iff \mu \in \partial f(x) \iff x \in \partial f^*(\mu).$$

  Gradient identities require differentiability at the corresponding points.

For example, the conjugate of $f(x) = 0$ on $\mathbb{R}$ is 0 at $\mu = 0$ and $+\infty$ elsewhere, illustrating why general differentiability cannot be asserted.

Examples.

- $f(x) = x^2/2$ on $\mathbb{R}$: $f^*(\mu) = \mu^2/2$.
- $f(x) = -\log x$ for $x > 0$: $f^*(\mu) = -1 - \log(-\mu)$ for $\mu < 0$, and $+\infty$ otherwise.
- $f(x) = x \log x$ on $x \geq 0$, with $0 \log 0 = 0$: the optimizer for finite μ is $x = e^{\mu-1}$, hence $f^*(\mu) = e^{\mu-1}$.

See Boyd and Vandenberghe, Section 3.3, for conjugacy rules and domains [2].

3.2 Convex Programs with linear constraints

Primal (Maximization):

$$\begin{aligned} \max \quad & \sum_i c_i x_i - f(x) \\ \text{s.t.} \quad & \sum_i a_{ij} x_i \leq b_j, \quad \forall j \\ & x_i \geq 0, \quad \forall i \end{aligned}$$

Dual (Minimization):

$$\begin{aligned} \min \quad & \sum_j b_j \lambda_j + f^*(\mu) \\ \text{s.t.} \quad & \sum_j a_{ij} \lambda_j \geq c_i - \mu_i, \quad \forall i \\ & \lambda_j \geq 0, \quad \forall j \end{aligned}$$

Interpretation & Insight. **Strong Duality & Slackness.** Under a suitable Slater constraint qualification and finite attained optima, primal and dual values agree. The solution is characterized by:

1. **Complementary Slackness:**

   $x_i > 0 \implies$ Dual Constraint is Tight, $\quad \lambda_j > 0 \implies$ Primal Constraint is Tight.

2. **Conjugate Pair:** $\mu \in \partial f(x)$ and $x \in \partial f^*(\mu)$; write $\mu = \nabla f(x)$ only when f is differentiable at x.

Alternatively, for a minimization primal, the forms are reversed:

Primal (Minimization):

$$\begin{aligned} \min \quad & \sum_i c_i x_i + f(x) \\ \text{s.t.} \quad & \sum_i a_{ij} x_i \geq b_j, \quad \forall j \\ & x_i \geq 0. \end{aligned}$$

Dual (Maximization):

$$\begin{aligned} \max \quad & \sum_j b_j \lambda_j - f^*(\mu) \\ \text{s.t.} \quad & \sum_j a_{ij} \lambda_j \leq c_i + \mu_i, \quad \forall i \\ & \lambda_j \geq 0. \end{aligned}$$

4 Fisher Markets: Eisenberg-Gale & Shmyrev

The Eisenberg-Gale (EG) program is fundamental in algorithmic game theory as it captures the equilibrium of linear Fisher markets, which maximizes budget-weighted log utilities, equivalently the weighted Nash product $\prod_i u_i^{B_i}$. Its dual variables provide the equilibrium prices.

4.1 The Eisenberg-Gale (EG) Program

lemma 4.1 – EG Duality

The EG Primal and its Dual are related as follows. The dual variables p_j correspond to item prices.

Primal (EG Program):

$$\begin{aligned} \max \quad & \sum_i B_i \log u_i \\ \text{s.t.} \quad & u_i \le \sum_j v_{ij} x_{ij}, \quad \forall i \\ & \sum_i x_{ij} \le 1, \quad \forall j \\ & x_{ij} \ge 0. \end{aligned}$$

Dual (Eq. 1):

$$\begin{aligned} \min \quad & \sum_j p_j - \sum_i B_i \log(\beta_i) \\ \text{s.t.} \quad & p_j \ge v_{ij}\beta_i, \quad \forall i, j \\ & p_j \ge 0, \quad \beta_i > 0. \end{aligned}$$

Remark. The displayed dual drops the additive constant $C_B = \sum_i (B_i \log B_i - B_i)$. At an optimum, $\beta_i = \min_{j: v_{ij}>0} p_j / v_{ij}$; this is the reciprocal of buyer i's maximum bang-per-buck.

Derivation and equilibrium interpretation. Introduce multipliers $\beta_i \ge 0$ for $u_i \le \sum_j v_{ij} x_{ij}$ and $p_j \ge 0$ for supply. Maximizing the Lagrangian over $x_{ij} \ge 0$ is finite precisely when $p_j \ge \beta_i v_{ij}$. Maximizing over $u_i > 0$ gives

$$\sup_{u_i>0} (B_i \log u_i - \beta_i u_i) = B_i \log B_i - B_i - B_i \log \beta_i,$$

with $\beta_i > 0$. This yields the displayed dual plus C_B. At an optimum, $u_i = B_i/\beta_i$ and complementary slackness gives $x_{ij} > 0 \Rightarrow p_j = \beta_i v_{ij}$. Thus each buyer uses only maximum bang-per-buck goods and spends $\sum_j p_j x_{ij} = \beta_i u_i = B_i$. Supply complementarity supplies market clearing. ■

4.2 Connection to Shmyrev's Program

We can derive the Shmyrev Convex Program by applying a change of variables to the EG Dual.

Interpretation & Insight. Logarithmic transformation and domains. Under the standing assumptions, equilibrium prices and β_i are positive. Only pairs with $v_{ij} > 0$ enter the logarithmic constraints; in the bidding program set $b_{ij} = 0$ on zero-valued pairs. To linearize the constraints in the EG Dual, we define:

$$q_j = \log p_j \quad \text{and} \quad \gamma_i = -\log \beta_i$$

The constraint $p_j \ge v_{ij}\beta_i$ transforms into $\log p_j \ge \log v_{ij} + \log \beta_i$, which becomes:

$$\gamma_i + q_j \ge \log v_{ij}$$

This transformation leads to a new pair of dual programs:

Transformed Dual (Eq. 2):

$$\begin{aligned} \min \quad & \sum_j e^{q_j} + \sum_i B_i \gamma_i \\ \text{s.t.} \quad & \gamma_i + q_j \geq \log v_{ij}, \quad \forall i, j. \end{aligned}$$

Shmyrev Program (CP):

$$\begin{aligned} \max \quad & \sum_{i,j} b_{ij} \log v_{ij} - \sum_j (p_j \log p_j - p_j) \\ \text{s.t.} \quad & \sum_i b_{ij} = p_j, \quad \forall j \\ & \sum_j b_{ij} = B_i, \quad \forall i \\ & b_{ij} \geq 0. \end{aligned}$$

The Shmyrev objective includes $+\sum_j p_j$; this is constant because $\sum_j p_j = \sum_i B_i$ on its feasible set. Consequently dropping this term leaves the same maximizers. The bids recover allocations by $x_{ij} = b_{ij}/p_j$ for positive prices.

4.3 Quasi-Linear Utilities

For markets where buyers have **quasi-linear utilities** (utility form: $\sum_j (v_{ij} - p_j) x_{ij}$), the standard EG Primal is hard to generalize directly. However, the *Dual* generalizes naturally.

lemma 4.2 – Quasi-Linear Duality

By adding the constraint $\beta_i \leq 1$ to the EG Dual, we capture quasi-linear utilities. The corresponding primal is then derived.

Dual (Modified EG Type):

$$\begin{aligned} \min \quad & \sum_j p_j - \sum_i B_i \log(\beta_i) \\ \text{s.t.} \quad & p_j \geq v_{ij} \beta_i, \quad \forall i, j \\ & p_j \geq 0, \quad 0 < \beta_i \leq 1, \quad \forall i. \end{aligned}$$

Primal (Derived for Quasi-Linear):

$$\begin{aligned} \max \quad & \sum_i B_i \log u_i - \sum_i m_i \\ \text{s.t.} \quad & u_i \leq \sum_j v_{ij} x_{ij} + m_i, \quad \forall i \\ & \sum_i x_{ij} \leq 1, \quad \forall j \\ & x_{ij}, m_i \geq 0. \end{aligned}$$

Here m_i represents retained money in the utility expression (up to the additive constant B_i). The penalty is summed over buyers. As above, the displayed dual omits C_B; it is equivalent up to an additive constant, rather than equal in objective value as written.

References

[1] N. R. Devanur, K. Jain, T. Mai, V. V. Vazirani, and S. Yazdanbod. *New Convex Programs for Fisher's Market Model and its Generalizations*. 2016. Author manuscript.

[2] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. Author's book page.